

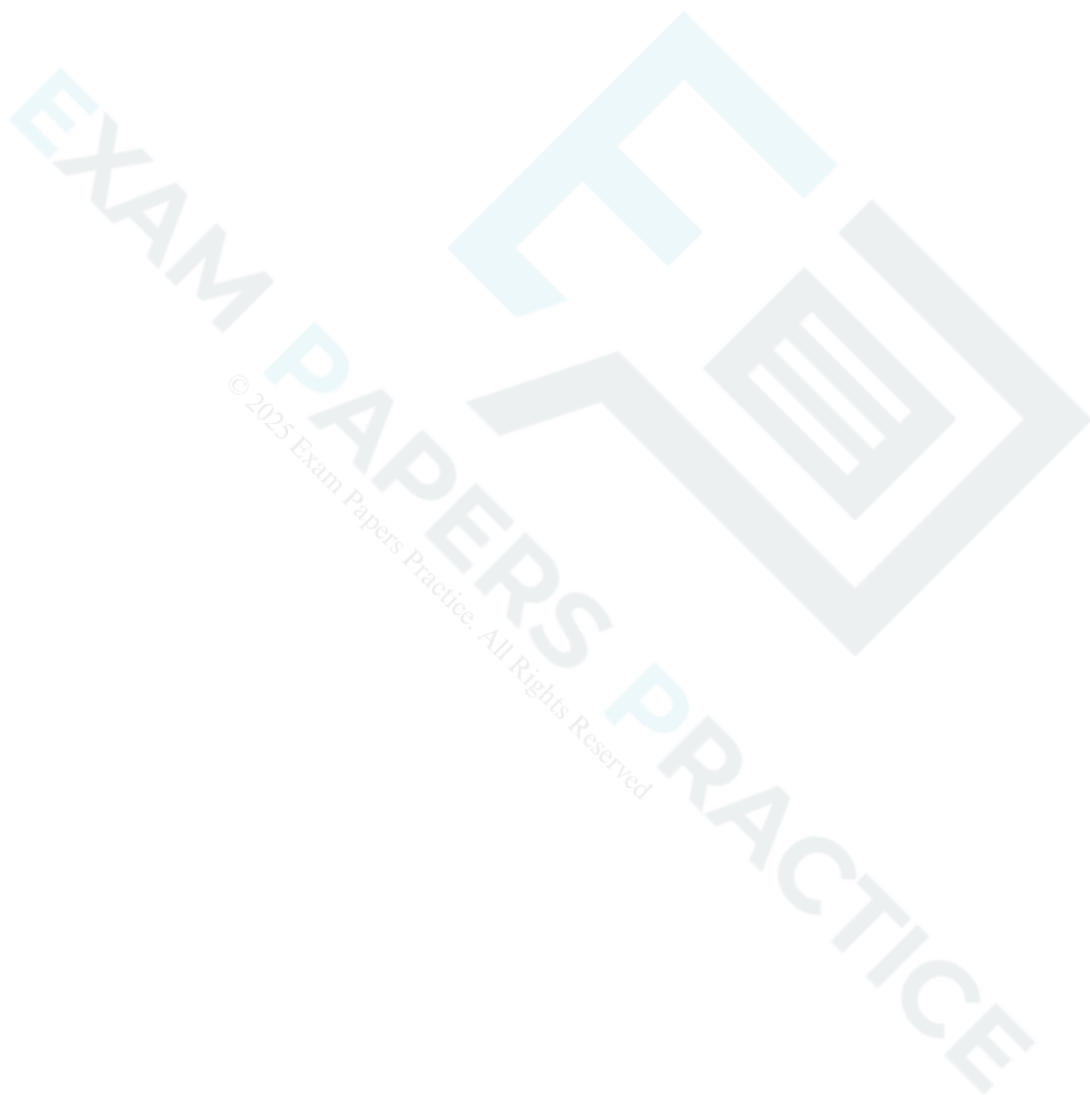
Mark Scheme

Q1.

Question Number	Answer	Mark
(a)	Application 2 Price elasticity of supply calculation % change in quantity supplied $(28\,268/134\,612 \times 100)$ $= 21\%$ (accept range 20.9 to 21) (1) $PES = 21/5.4$ (1) $= 3.9$ Award 2 marks for 3.9 (Accept a range between 3.8 and 4)	(2)

Question Number	Answer	Mark
(b)	Application 1 The only correct answer is C A is not correct because the PES does not equal to infinity B is not correct because the PES does not equal zero D is not correct because the PES is not less than one	(1)

Question Number	Answer	Mark
(c)	Knowledge 1, Analysis 1 Identification of a factor determining PES (1) e.g. • levels of spare capacity • stocks of finished goods and components available e.g. bricks • time period and production speed • level of factor mobility/flexibility • availability of bricklayers, electricians etc • availability of land • planning permission/regulations • availability of technology/machinery to build Explain as to the impact on price elasticity of supply (1) e.g. • limited availability of land/builders (1) so supply will be inelastic (1)	(2)



Q2.

Question Number	Answer	Mark																								
(a)	Application 2, Knowledge 2 Application: up to 2 marks for completing the table accurately. 1 mark for any correct answer in the column and 2 for all accurately complete/completed to the appropriate point. <table><tr><th>Price</th><th>Quantity demanded per month (000s)</th><th>Quantity supplied per month (000s)</th><th>New quantity supplied per month (000s)</th></tr><tr><td>£25</td><td>400</td><td>320</td><td>240</td></tr><tr><td>£30</td><td>360</td><td>360</td><td>280</td></tr><tr><td>£35</td><td>320</td><td>400</td><td>320</td></tr><tr><td>£40</td><td>280</td><td>440</td><td>360</td></tr><tr><td>£45</td><td>240</td><td>480</td><td>400</td></tr></table> Knowledge/understanding: 1 mark for identifying the original and new equilibrium (1) Current equilibrium price = £30 New equilibrium identified = £35 (where QD = 320 and QS = 320) (1) 1 mark for calculating the change in the equilibrium price (1) £35-£30 = £5 increase	Price	Quantity demanded per month (000s)	Quantity supplied per month (000s)	New quantity supplied per month (000s)	£25	400	320	240	£30	360	360	280	£35	320	400	320	£40	280	440	360	£45	240	480	400	
Price	Quantity demanded per month (000s)	Quantity supplied per month (000s)	New quantity supplied per month (000s)																							
£25	400	320	240																							
£30	360	360	280																							
£35	320	400	320																							
£40	280	440	360																							
£45	240	480	400																							

(4)

Question Number	Answer	Mark
(b)	A	(1)

Q3.

Question Number		Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge 1 mark for identification of a possible cause and 1 mark for its relevance to energy use Possible causes: • Increase in world population (1) • Increase in devices requiring electricity (1) • Increase in world GDP and living standards (1) Application: 2 marks for data references from Figure 1 (identifying the forecast change): • Demand for nuclear and renewable more than double • Demand for gas doubles • Demand for coal and oil increase by between 30% and 50% • Comment on increase in overall demand Analysis: • 1 mark for explanation of the link between the identified case and energy use	(5)

Question Number		Mark
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge and analysis • Increasing injections (exports) (K) may lead to a boost to the circular flow and an increase in AD if not offset by greater leakages (A) • Decreasing leakages (imports) (K) may lead to a reduction in the circular flow and reduce AD (A) Application: • Exports increasing, imports decreasing, but until 2020, leakages still greater than injections so overall leakage from circular flow (2) • Or 2 specific data references from figure 4 Evaluation (2 marks for any relevant point): • These are forecasts so the impact on the circular flow may be different from that described above • Other injections and leakages influence the circular flow which may offset the changes described above	(8)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 • recognition that gas and solar energy/nuclear energy are substitutes • application of cross elasticity of demand: definition or formula value will be positive • use of information provided to draw conclusion (reference to first paragraph, Extract 2): a fall in the price of gas will lead to a fall in the demand for solar energy and nuclear energy.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 • much will depend on relative price of gas in relation to solar and nuclear energy • recognition that there will be a time lag before any impact on market for other forms of energy because supply very price inelastic • investors will want to be certain that change in relative prices is long term • impact may be limited due to the rising demand for global energy	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic effects: • on price of gas to consumers – impact on consumers’ surplus • on price of gas to producers – impact on producers’ surplus • implications for producers of other forms of energy • external costs – on environment and water supply Macroeconomic effects: • on investment – multiplier effect on GDP so promoting economic growth • on employment – an estimate that 35 000 jobs will be created • balance of payments on current account e.g. competitiveness may increase if fracking results in lower energy prices leading to an improvement in exports. Also, less need to import energy from abroad • rate of inflation may fall – lower energy prices Possible evaluation points include: • Impact on price of energy may be less significant than in the USA • Impact on other forms of energy – much will depend on how much gas is obtained from fracking • Difficult to forecast the magnitude of external costs or to place a monetary value on them • Impact on macroeconomic variables (growth, employment, balance of payments) dependent on amount of gas found, the price at which it sells, changes in prices of other forms of energy	(25)

Knowledge, application and analysis		
Level	Marks	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on a small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

Evaluation		
Level	Marks	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning. Evaluation recognises different viewpoints and can be critical of the evidence provided.

Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8 Microeconomic effects: • impact on revenues and profits of energy companies • impact on investment decisions by firms in the energy market • impact on wages of workers in energy industries • impact on consumer real incomes and on demand for other goods and services. Macroeconomic effects: • lower costs of production • lower global rates of inflation (AD/AS analysis) • impact on global growth rates • impact on balance of payments on current account of different countries • increase in CO₂ emissions: consideration of impact on climate change on different countries, reference to external costs • impact on consumers and producers of the main countries supplying energy. Possible evaluation points include: • impact on different industries will depend on their energy costs as a proportion of total costs • impact on households will depend on cost of energy in household budgets • impact on demand for other goods and services will depend on other factors e.g. state of world economy, consumer confidence, household debt • impact on inflation rates and growth rates depends on elasticities of AD and AS curves • impact on balance of payments on current account of different countries depending on whether they are net importers or net exporters of energy • impact on external costs depends on how different energy sources are affected	(25)

Knowledge, application and analysis		
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	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on a small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models
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Level 3	7–9	Evaluative comments supported by relevant reasoning. Evaluation recognises different viewpoints and can be critical of the evidence provided.

Q4.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge: 2 marks for identification of how intervention might work e.g. • Raising/lowering interest rates (1) which affects exchange rates/hot money flows (1) • Buying/selling foreign currencies (1) which affects their exchange rate (1) • Buying/selling domestic currencies (1) which affects their exchange rate (1) • Quantitative easing/changing the money supply might be used (1) • Sets a ceiling and floor (1) Application: 2 marks for data references e.g. • Use of Figure 1 to show interest rate changes: e.g. central bank rose rates to 6.25 percentage points in 2018 (1) which caused the lira to appreciate 11% (1) lira has lost a third of its value against the dollar in 2018 (1) • Use of text to give example of Turkish government buying and selling: e.g. the country is highly indebted on foreign currency (1) • This might be shown by an exchange rate diagram (1)	
	Analysis: 1 mark for explanation e.g. • Process by which the intervention affects exchange rate/interest rate e.g. the government buys US dollars to decrease the price of Turkish lira (1) by increasing the supply of lira on the forex market (1) or raises interest rate to increase the price of Turkish lira (1) NB a diagram can be seen as application or analysis, if used in a valid context.	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge/ implicit understanding: 2 marks (1+1) for each reason and explanation e.g. a lower exchange rate can: • Reduces financial risks • To bring stability/security in the economy • Harms borrowers who have borrowed in foreign currency • Causes inflation as import costs rise • Worsens the terms of trade • Worsens living standards • Short term J-curve effects on current account of the Balance of Payments Application: (2 marks: 1+1 or 2) • Turkish inflation 25% • Lira has lost a third of its value of US dollar 2018 (1) • Turkey is highly indebted in foreign currency (1) • Government takes measures against excessive appreciation or depreciation of the Turkish lira to reduce financial stability risks (1) • The financial market faces difficulties in trying to restore <i>foreign investors'</i> confidence (1)	

	Analysis: 2 marks (1+1) for linked development of each point e.g. • A weaker currency can lead to cost push inflation as import prices rise • A weaker currency worsens the terms of trade as export prices fall relative to import prices • There will be less investment if there is a loss of foreign investors' confidence/AD shift Evaluation (2 marks: two points 1+1 or 2). Points might include: • Inflation is very high (25%) so this should be main reason not to allow a fall in the exchange rate • Inflation is showing signs of reversing so a fall in the exchange rate may be less problematic • Problems caused by inflation might be outweighed by the other benefits such as export-led growth/improved competitiveness • Benefits of weak exchange rate, e.g. exports cheap, • Other benefits of weak currency e.g. stimulus of economy, increased competitiveness, higher employment and FDI	(8)
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Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 Knowledge/implicit understanding: Policies identified might include: • Indirect tax/charges • Ban or other regulation • Free bags/nets or other state provision • Fines • Behavioural economics policy Application: • Tax or charge on each item. Might be illustrated through a specific tax diagram or similar. • Ban in some parts of Turkey for specific types (carcinogenic black bags). Might be supported by a limit on supply diagram. • Fines for consumers and retailers in some parts of Turkey • Free bags/nets or other state provision in e.g. 20 000 in Corlu near Istanbul • 950 tonnes of plastic waste Analysis: • Explanation of process linking the intervention to the reduction in use of bags. • This might be shown through a diagram(s).	(8)

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	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 - Charges work in other countries, e.g. UK - Emerging economies/countries might have more problems facing the cost of interventions e.g. inequality issue, poverty - The problem is much wider than plastic/single use bags, and the interventions will only affect a few on-street purchases - Likely to succeed because it taps into the social conscience of consumers and even a small charge may have a dramatic impact as in the UK - The tax/charge makes up a small proportion of income so many have minimal impact. PED is very inelastic/close to zero - Unintended consequences: e.g. multiple-use/ bags for life often contain more plastic and unless used many times can be more environmentally damaging - First paragraph suggests they are everywhere so given habitual behaviour people may continue to use them despite charges/ tax - opportunity cost of giving away 20 000 bags in a month – is this socially efficient?	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 16 marks for KAA, for effects of increased interest rates. Microeconomic effects may include: • Fixed costs to firms will rise. May cause shut down/loss of competitiveness • Fall in consumer demand as consumption expenditure falls - likely reduction in AR for normal goods • Other costs to consumers/firms/a government e.g. increased burden of debt and loan repayments/servicing costs • Fall in investment by firms - reducing dynamic efficiency in the economy as less funds for R&D • Impact on Turkish housing market Macroeconomic effects may include: • Link between higher interest rates and lower investment and lower consumption- reducing AD: inward shift of AD (or slowing outward shift) with multiplier effects with macroeconomic impacts e.g. unemployment • Rise in the value of the lira – worsening of current account of the Balance of Payments – possibly with effects on other countries • Redistribution of income e.g. from borrowers to savers/lenders, from asset holders to renters • Cost of financing public deficit/debt (may refer to opportunity cost) may mean less government spending elsewhere, e.g. limited ability to spend on education NB for a Level 4 response there must be micro and macro effect (s). Income distribution, investment changes and competitiveness, for example, can be used as either micro or macroeconomics.	

	9 marks for evaluation – points might include: • Interest rate might not be the most significant cost for firms • Contractionary monetary policies already very tight. Depends on the interest rates in other markets • Inflation has peaked and there are signs that it will fall so the rise in interest rates is not necessary • Trade-off between objectives or prioritisation e.g. lower inflation at the cost of higher unemployment might be shown with critical use of a Phillips Curve • Depends on interest rates in other markets- do we mean lending rates in other countries. • Turkey's private sector is resilient - they may refer to this explaining that shut down is less likely • Impact on housing market will depend on whether people own homes and are mortgaged or whether they rent, or whether mortgage is fixed or variable rate • Magnitude the interest rate change - Figure 1 suggests large increases and therefore significant impact/large changes are unable to have much further effect	(25)
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Knowledge, application and analysis		
Level	Mark	Descriptor
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Level 1	1–4	Displays ability to apply knowledge in context but will focus on small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models.
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis.
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
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Evaluation		
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Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 16 marks for KAA, for effects of increased infrastructure spending Microeconomic effects may include: • Wages higher • Improved infrastructure affects firms e.g. businesses' fixed or variable costs fall as transport costs fall • Investment seen to boost business confidence, lead to further investment and spending within firms • External benefits/external costs associated with the project - especially given the size, number of runways and amount of traffic outlined in Extract C. • External economies of scale • Accept answers based on impact on costs and/or revenues if the building occurs in the private sector, with appropriate diagram Macroeconomic effects may include: • Redistribution of incomes • Outward shift of AD/injection (I or G) into circular flow (or further outward shift in AD) with multiplier effects, with macroeconomic impacts e.g. employment and incomes rising • AS shift right with a movement in capacity of the productive economy shown (Keynesian or Classical AS) • Fiscal loosening – effects of increased budget deficit if funded without increase in tax • Effects of a rise in tax, if linked to government decision to raise tax to fund the projects	

	NB for a Level 4 response there must be micro and macro effect (s). Some factors could be seen as micro or macroeconomics, such as a rise in tax, increased FDI, redistribution and confidence. Up to 9 marks for evaluation. Points might include: • Opportunity cost of spending on infrastructure e.g. schools, or higher taxes, • Crowding out – financial or resource • Might be a white elephant project • Depends on relations with EU/other countries as to whether the airport will reach full capacity • Depends on the size of the multiplier effect • Depends on the state of the financial sector • True costs not accounted for - externalities ignored • Large projects require a lot of coordination and getting behind schedule putting lives at risk. • Difficulty in placing value on external benefits and costs associated with the project	(25)
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Knowledge, application and analysis		
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Level 2	4-6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7-9	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Q5.

Question Number		Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge and Analysis - Definition of exchange rate: price of one currency in terms of another (1) - Possible reasons: - Reduction in QE by USA (K): likely to cause a rise in interest rates in developed economies and less demand for the currencies of emerging economies (A) - Slow-down in the Turkish economy (K): likely to cause a fall in confidence and a decrease in foreign direct investment and less demand for the currencies of emerging economies (A) Application: Reference from Figure 1: (2) 14% fall in value of the Turkish lira over the whole period	(5)

Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 4 • understanding of <i>inflation</i>: general increase in the average price level Costs of inflation include: • arbitrary redistribution of income from lenders to borrowers • loss of international competitiveness of Argentine goods • uncertainty causing a fall in investment • industrial unrest: reference to impact of strike by the police • menu costs and shoe leather costs • danger of hyperinflation if the current inflation is not brought under control.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(b) continued	Evaluation 4 - current inflation rate may be a short-term phenomenon only - impact of competitiveness depends on inflation rates of countries that Argentina trades with - Argentine government is taking measures e.g. price controls to reduce the rate of inflation - impact on living standards depends on what is happening to wages	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number		Mark
(c)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge and Analysis - Restrictions on beef exports caused a fall in demand and in the price of beef (K). In turn, this made it less profitable to produce beef. (A) (2) - Soya production becoming more profitable because of growing demand from China (K). Therefore, farmers reallocating resources from beef production to soya production. (A) (2) Application 10 million fall in the stock of cattle (over 10 years) to 48 million in 2011. - Exports accounted for 20% of beef production in 2007, but have fallen to just 7% in 2013 (2) Evaluation 2 marks for any valid point: - Some farmers may find it difficult to switch production from beef to soya because of the costs involved and/or unsuitability of the land - Farmers might think that government restrictions on beef exports may be short-term so they will not switch from beef to soya.	(8)

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8 Evaluation 9 Microeconomic effects: • supply and demand diagram illustrating the impact of maximum prices: contraction in supply and extension of domestic demand • shortages of products subject to price controls • need for rationing • greater price stability for producers • implications for resource allocation: producers will switch resources to the production of products not covered by price controls. Macroeconomic effects: • falling export revenues from beef sales • negative impact on Argentina's balance of trade • could cause fall in the rate of economic growth • negative implications for Argentina's economic development • fall in the rate of inflation • price controls might act as an incentive for producers to diversify and/or develop manufacturing and tertiary industries. Possible evaluation points include: • impact of price controls depends on how rigorously they are enforced • producers may not reallocate resources to products not subject to price controls if they believe that they will only exist for a short time • impact on the balance of trade may be minimal, e.g. because of the increase in exports of soya • impact on the rate of inflation may be limited to the short-term: once price controls are removed the rate of inflation might increase significantly	(25)

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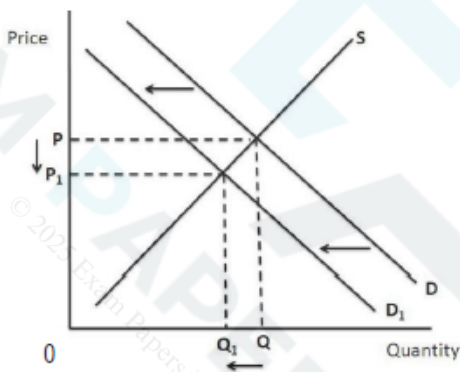
Evaluation		
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Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic effects: • primary producers: export revenues may fall since demand for many primary products is price inelastic • increase in costs for firms reliant on imported raw materials • increase in competitiveness for Argentine exporters of manufactured goods. Macroeconomic effects: • increase in the competitiveness of Argentine's goods and services may lead to an improvement in the current account of the balance of payments • increase in the rate of economic growth • leading to an increase in employment • cost push inflation resulting from increased costs of imported goods • increase in the real burden of foreign debt. Possible evaluation points include: • impact on individual Argentine producers depends on their reliance on imports and/or exports • demand for some products, e.g. soya may not be adversely affected because demand is increasing rapidly in China • inflation is very high so any competitive advantage may be lost quickly. Impact on current account will depend on whether the Marshall-Lerner condition is fulfilled • short-run and long-run effect on the current account may be analysed using the J curve effect • the current account will only improve if the Marshall-Lerner condition is fulfilled, i.e. if the sum of the price elasticity of demands for imports and exports is greater than 1	(25)

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Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning. Evaluation recognises different viewpoints and can be critical of the evidence provided.

Q6.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 2, Analysis 2 Knowledge 1 mark for definition of revenue or may identify clearly in diagram (1) Application 1 mark for using source data or linking ban to reduced consumption (1) and 1 mark for diagrammatic link to change in revenue (1), e.g. • "can reduce tobacco consumption" (1) leading to decreased revenue as shown in diagram (1) Analysis Diagram showing demand shifts left or becomes less price inelastic (1) and new equilibrium (1)  (5)	
Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 2 • Application – price of e-cigarettes per nicotine cheaper • Data on PED suggests some rational response to price change e.g. 10% rise causes 4% fall in demand (Extract A) Reasons rational • Smokers computationally aware of financial gains of switching • Smokers informed of private health gains (Extract B Paragraph 2) • Reducing health external costs e.g. passive smoking (Extract A L15) • Smokers aware that regulations will increasingly make it difficult for them to gain utility of nicotine consumption through tobacco smoking NB Reasons for/against being rational may be presented as either KAA or EV (6)	

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(b) continued	Evaluation 4 Reasons against being rational • Smokers lack information and computational skills to accurately assess full costs and benefits of switching. May underestimate long-term gains of switching (bounded rationality) • Information overload • Both products contain nicotine which is addictive and habit forming. • If e-cigarettes were not available they may quit smoking altogether • Switch may be only short-term (impulse buying) • E-cigarettes complements tobacco smoking resulting in an overall increase in nicotine consumption • PED for tobacco remains lower than e-cigarettes so a price rise in the latter may result in a switch back • Peer pressure (herd mentality) pushing smokers to make the switch	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Answer	Mark
(c)	Knowledge 2, Application 2, Analysis 2 Knowledge/understanding 1 mark each for any two of the following: • definition or formula of cross elasticity of demand (1) • tobacco cigarettes and e-cigarettes have a positive cross elasticity of demand (1) • understanding of substitute relationship e.g. similar products (1) Application 1 mark each for any two of the following: • e-cigarettes seen as less harmful so become more attractive compared to tobacco cigarettes (1) • tougher regulations and growing health awareness on tobacco (1) • they are substitutes because they offer nicotine (1) • applying cigarettes to diagram showing impact on e-cigarettes (1+1) • relative price differences e.g. Figure 1 or tax rises on tobacco (1) • use of numerical example e.g. a 10% rise in the price of tobacco cigarettes results in a less than 10% rise in demand for e-cigarettes Analysis Up to 2 marks for any one of the following in linked development OR two limited examples of analysis (1+1): • the increase in the price of traditional cigarettes is resulting in the change in quantity demanded of e-cigarettes increasing • consideration of whether distant or close substitutes • diagram showing relationship between price of one of the product and demand for the other • magnitude/time	(6)

Question Number	Answer	Mark
(d)	Knowledge 2, Application 2 2 marks for each correct answer, e.g. • High-income countries = -4% / $+10\%$ (1) $= -0.4$ (2) or 0.4 (2) • Low-middle-income countries = -5% / $+10\%$ (1) $= -0.5$ (2) or 0.5 (2) NB Award 1 mark for a correct formula as an alternative for substituting correct figures on up to one occasion.	(4)

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 • Identification of determinants of price elasticity of demand • Application – PED tobacco cigarettes -0.35; PED e-cigarettes -1.9. Reasons why tobacco cigarettes being relatively price-inelastic, e-cigarettes relatively price-elastic e.g.: • habitual consumption • tobacco broadly defined • proportion of income spent on tobacco cigarettes lower than switching cost of buying reusable e-cigarette kit • number of close substitutes • tobacco addictive necessity, e-cigarettes luxury (not yet addicted to). NB to access Level 3 candidates need to consider tobacco and e cigarettes.	(9)

"Assess" was used as the command word for this question instead of "Discuss". In order to meet the expectations set out in the sample material the 10 mark level descriptors have been mapped on to the 15 mark levels as below on this one occasion. See Appendix 1 for full details.

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	4-6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	7-9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • significance of determinant • e-cigarettes habit forming • tobacco cigarettes narrowly defined • proportion of income spent on tobacco cigarettes is rising • becoming closer substitutes • tobacco addiction waning as lost glamour and health warnings, e-cigarettes nicotine addiction. • longer term response to rising tobacco prices	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Indicative content	Mark
(f)	Knowledge 4, Application 4, Analysis 6 Functions of the price mechanism to allocate resources: • rationing • incentive • signalling Market diagrams for tobacco and e-cigarettes, making use of supply and demand analysis. * Consumers with knowledge of the potential harm of cigarette smoking may quit smoking. * Smokers will maximise their private utility * Consumers will decide what is in their best interests * Reduction in smoking will happen automatically, without government intervention, as e-cigarettes become popular.	

	Candidates may refer to Adam Smith Consumers free to pursue own self interest Profit incentive will mean firms will want their consumers to live long – so move out of tobacco and into e-cigarette market. Without government regulations e-cigarettes market will be easier for new firms to enter thus improving competition. Competition between e-cigarette manufacturers to maximise profits will drive health improvements and lower costs making them more price competitive. Candidates may refer to Friedrich Hayek Resource allocation from tobacco to e-cigarettes will be brought about by market forces, which will be far superior to any state planning system. Individuals do not have perfect information on cigarettes but have the best knowledge of their own situation. The price mechanism acts as a communication network, markets are self-correcting. Government lacks sufficient information about everything happening in the cigarettes market to be able to make informed decisions.	(14)
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Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(f) continued	Evaluation 6 Failings of the price mechanism • Information gaps • Asymmetric information • Externalities • Peer pressure ignores utility • Habit forming distorts utility Possible different disadvantages or approaches : * While consumers may maximise their private utility $MSC > MSB$, with explanation. e.g. passive smoking/health care costs * Without government intervention the oligopolistic/monopolistic power of cigarette firms, in the form of advertising and propaganda, will go unchecked Candidates may refer to Karl Marx Consumers exploited in past by misinformation on tobacco and due to being addictive they become dependent. Competition and the need to profit maximise will mean firms cut costs risking consumers health and increase prices thus reducing consumer surplus. Nature of free market is for powerful monopolies to emerge that can further exploit consumers.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Question Number	Indicative content	Mark
(g)	Knowledge 4, Application 4, Analysis 6 Positive/negative effects of government intervention on tobacco cigarettes and or e-cigarettes with reference to market failure and using diagrams in various contexts: • Indirect taxation (ad valorem and specific) • Subsidies for tobacco replacements • Minimum cigarette price • Provision of information e.g. public health campaign • Regulation e.g. banning smoking in public places or advertising. • Direct provision – nicotine patches and e-cigarettes on NHS prescription Diagrams may be used to illustrate the above.	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(g) continued	Evaluation 6 Negative or unsuccessful impact of the intervention methods. Understanding of government failure as intervention that results in a net welfare loss – tobacco cigarette tax revenue exceeding tobacco related NHS expenditure. Causes of government failure: • Distortion of price signals – harm reduction instead of prevention, consuming e-cigarettes rather than quitting • Unintended consequences – e-cigarettes draw in new users acting as a gateway to smoking; regulating e-cigarettes protect tobacco from competition; tobacco smuggling • Excessive administrative costs – slow down e-cigarette innovation • Information gaps – little research on long term effects of e-cigarettes • Ineffective as $PED < 1$; diminishing returns of tax and regulations. Price inelastic demand diagram. Other evaluation approaches include: • Consideration of the extent of government intervention e.g. could it be more regulated • Free market approach might be more effective • Depends on the country being considered • Short run and long run	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Appendix 1

Part (e) directed candidates to "Assess" for 15 marks. In order to be in line with the support materials for this specification, this question should instead have directed candidates to "Discuss" for 15 marks. In order to ensure that all candidates received the fairest possible outcome, examiners were briefed on a levels-based marking structure for both command words. The body of the mark scheme includes the levels structure for the "Assess" command word as written in the paper. The levels structure that examiners considered for the "Discuss" command word is included below.

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 • Identification of determinants of price elasticity of demand • Application – PED tobacco cigarettes -0.35; PED e-cigarettes -1.9. Reasons why tobacco cigarettes being relatively price-inelastic, e-cigarettes relatively price-elastic e.g.: • habitual consumption • tobacco broadly defined • proportion of income spent on tobacco cigarettes lower than switching cost of buying reusable e-cigarette kit • number of close substitutes • tobacco addictive necessity, e-cigarettes luxury (not yet addicted to). NB to access Level 3 candidates need to consider tobacco and e cigarettes.	(9)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • significance of determinant • e-cigarettes habit forming • tobacco cigarettes narrowly defined • proportion of income spent on tobacco cigarettes is rising • becoming closer substitutes • tobacco addiction waning as lost glamour and health warnings, e-cigarettes nicotine addiction. • longer term response to rising tobacco prices	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Q7.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge and analysis Up to 3 marks for: - the meaning of externalities, e.g. effect on third party, outside the transaction, difference between social cost and private cost (1). Points might include: - use of plastic which is hard to recycle/other single use plastic arguments (1) - HFSS food can cause obesity and have third party effects (1) - more congestion/air pollution from food deliveries (1) (positive) externality due to rise in productivity generated by cash rich, time poor professionals might allow people do impact third parties positively (1) Application: 2 marks for data (1+1 or 2): 1.4 billion containers (1) or other plastic items to allow food to travel safely e.g. heat retention (1) - over-consumption / order more food (1) e.g. impact on other NHS users (1) - HFSS food (1) - falling profit margins (1) or increased competition in food delivery might mean wages fall so fewer wider macro benefits (1) - don't even have to get up/lack of exercise of consumers ordering from their sofa (1) NB 2 marks may be awarded for a diagram e.g. a negative externalities in production of food that has to be delivered	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge/implicit understanding: (1+1) marks e.g. • demand is price inelastic means that the percentage change in price causes a smaller percentage change in quantity demanded • demand is income elastic means that percentage change in income causes a greater percentage change in demand Application: (1+1 or 2) e.g. • food delivery firms' charge of £5 is a very small percentage of consumer income so they might be seen as essential or basic goods (Ext B line 5) • three dominant firms Just Eat, Uber Eats and Deliveroo (Ext B line 2) • The food delivery industry has boomed in recent years (Ext B line 2) • cash rich, time poor (Ext B line 7) • income growth prior to 2020 meant delivery has boomed • HFSS addiction/habit forming • Convenient mobile app, so PED inelastic • 5 times more expensive than cooking at home Analysis: (1+1) development of each type of elasticity, e.g. • if demand is price inelastic then revenue will rise for the firm when it raises price • habit forming behaviour so PED inelastic • few close substitutes so PED inelastic • small part of overall consumer spending so PED inelastic • food delivery services are a very small percentage of consumer income so they might be seen as essential or basic goods • if food delivery is a luxury the demand likely to be income elastic, e.g. then the firms will benefit in times of economic growth, but lose demand in a recession • if food delivery is a treat in an economic downturn then it is an inferior good NB up to 2 marks may be awarded for diagram(s) (1+1)	
	Evaluation: (2 marks for any relevant point, or two points 1 +1) • the 30% of prices paid by restaurants is not seen by the consumer • depends on geographical areas e.g. rural vs urban areas. • the worldwide health crisis might have caused distortions in the data • third-party food delivery firms are inferior goods if demand rises when incomes fall • other factors affect demand apart from income, e.g., availability of substitutes (eating in restaurants is impossible in global health crisis) • discussion of normal vs inferior question e.g. compared to eating in a restaurant or cooking at home • income elastic vs. inelastic e.g. demand for third-party food delivery is price elastic for some income groups when incomes fall as it takes up a larger percentage of income	(8)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 Points might include: • integration includes horizontal/vertical/ conglomerate joining of firms • leads to economies of scale • increased market share • removal of competition • increased monopoly power • international rivals can avoid taxes • prices higher for consumers • reduced choice for restaurants and consumers • monopsony power over restaurants • technical advantages of vertical integration • vertical power over firms can leave other firms with high barriers to entry • Just Eat has bought out rivals e.g. merger between Just Eat and Takeaway.com • Amazon has partially bought Deliveroo (16% so far) which may have saved it, and/or improved it on the tech side NB effects might include more than one types of economies of scale, or other combined analysis. Award diagrammatic analysis e.g. economies of scale LRAC	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.

Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.
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Question Number	Indicative content	Mark
(c) continued	Evaluation 4 • mergers might be necessary to ensure the survival of firms (e.g. Deliveroo case) • further mergers might be limited by the CMA • supernormal profits might lead to lower prices, better technology and/or a more consumer-led market • prices may fall due to fall in average costs due to e.g. economies of scale • diseconomies of scale	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic effects of depreciation may include: • higher cost of imports of food and therefore higher cost of production for hospitality sector • lower profits for firms in hospitality • higher prices of food for consumers • consumers have lower standards of living if they are on fixed incomes/ • falling real incomes so demand for restaurant meals and food delivery services might shift left • workers have higher incomes if working in exporting firms in other industries, so might use more delivery services • restaurants might begin to use more locally produced food Macroeconomic effects of depreciation may include: • capital in-flows, e.g. benefits to multinational firms such as Just Eat for investment in the US using other currencies/ AD and/or LRAS shift out • cost push inflation/SRAS shifts in • more injections and fewer withdrawals leading to rise in AD and multiplier effect • increased employment • incentive to use transfer pricing • UK tourism might be boosted by depreciation which would improve local restaurants NB for a Level 4 response there must be micro and macro effects and use of data with reference to restaurants or other food delivery services. Higher costs, changes in competitiveness, environmental changes could be seen as micro or macro, for example, can be used as either micro or macroeconomics. NB Effects in KAA could be used as evaluation and vice-versa.	(25)

	9 marks for evaluation – points might include: • depends on the nature of the restaurant. Some try to use locally sourced ingredients • the pound has recovered significantly since 2016 • food costs are a small cost relative to other costs, e.g. rent and delivery • some restaurants are specialised in UK food so cost effect very low • other factors are more important, e.g. the heavy fiscal spending in response to the global health crisis • other factors determining capital inflows • relative depreciation or trade weighting of other currencies • offsetting costs with altering portion sizes or changing menu options • impact on tourism small relative to the increase in the production of food • regional differences with the UK e.g. whether there is a local food supply option • Marshall Lerner • J-curve	
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Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models.
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis.
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

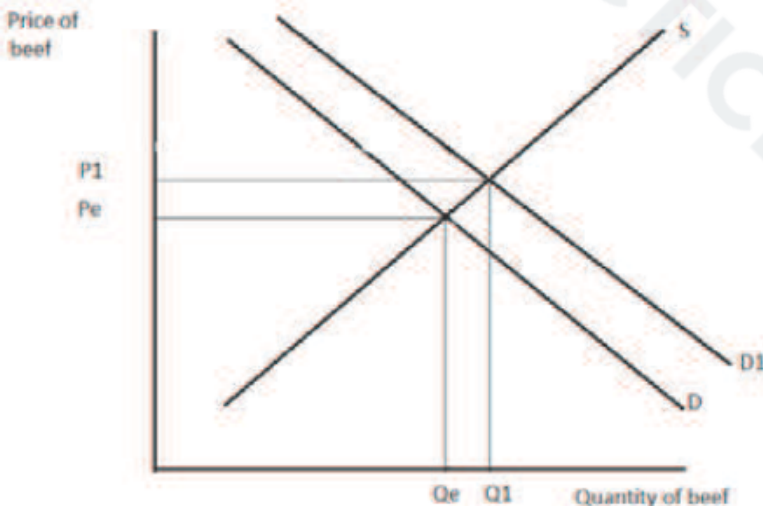
Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 16 marks for KAA, for effects of increased minimum wage Microeconomic effects may include: • rise in wages will benefit very low paid workers. Might be shown on increasing NMW diagram • effect on cost to firms e.g. reduced profit margins. Might be shown on a cost and revenue diagram. • excess supply of hospitality workers • effect on prices consumer surplus falls, output falls, price increases • impact on quantity of labour demanded/contraction of labour demand • impact on productivity • inefficient resource allocation due to distortion of price mechanism Macroeconomic effects may include: • income equality improvements – may use Gini/Lorenz • prices rise – inflationary effects • reduction in relative poverty • AD: increase in consumption as incomes increase/fall in investment as profits fall • SRAS: cost-push • LRAS: if investment falls, then LRAS shifts in • reduced employment • higher income tax revenue • lower unemployment related benefits NB for a Level 4 response there must be micro and macro effect (s) and use of data with reference to restaurants or other food delivery services. Employment or welfare issues could be seen as micro or macro, for example, can be used as either micro or macroeconomics. NB Effects in KAA could be used as evaluation and vice-versa.	(25)

	9 marks for evaluation – points might include: • effect on firms depends on how much wage costs are as a proportion of overall costs • impact on real output depends on level of spare capacity/elasticity of LRAS/magnitude of multiplier • effect on workforce of employment regulation depends on elasticity for demand for labour, and degree with which it can be replaced with capital • depends on PED of food delivery firm market as a whole • consumers might also have increases in wages so profits might not fall • some firms will face decreased demand and go out of business. Loss of jobs and other economic activity • depends on the PED and PES in factor market and product market as a whole • LRAS might rise as economically inactive workers enter labour force to look for jobs • higher tax revenue may not occur if unemployment increases or investment decreases • higher unemployment related benefits if unemployment increase • productivity/motivation arguments	
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Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models.
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis.
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Q8.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 2, Analysis 2 Knowledge 1 mark for original supply, demand and equilibrium. Application 1 mark for using source data (1) and 1 mark for link to change in demand (1), e.g. - increased meat sourced from UK, 81% to 83%/increased sales to emerging markets (1) leading to increased demand (1). Analysis - Diagram – showing shift in demand (1) and new equilibrium (1). 	(5)

Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 2 - Define cross elasticity of demand. - Identify lamb as substitute. - XED – positive – as price of beef rises the demand for lamb will rise. - Diagram to show increased demand for lamb. - Price and quantity both rise for this good. - Likely to lead to increased revenue. - Increased producer surplus. - Increased consumer surplus.	(6)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no links between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relate them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(b) continued	Evaluation 4 • Magnitude – depends on amount of price adjustment. • Short run little change – people do not adjust to price changes straight away/long run may take time for people to substitute. • Closeness of substitutes – inelastic supply means little impact/elastic larger impact.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Answer	Mark
(c)	Application 2, Knowledge 2, Analysis 2 Application 2 marks for, e.g. • desire for protein/dairy – 30–50m litres (1) • increased demand (1). Knowledge/understanding e.g, any two from: • higher prices (1) • higher output (1) • revenue rise (1). Analysis e.g, any two from: • increased profits/producer surplus identified/shown in diagram (1) • investment required for machinery and logistics – getting products to China (1) • firms will export more products (1).	(6)

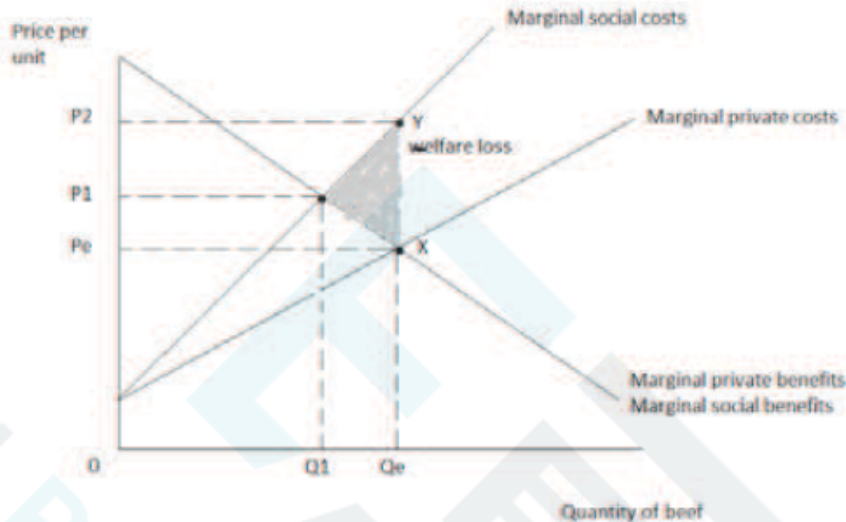
Question Number	Answer	Mark
(d)	Knowledge 2, Application 2 2 marks for correct use of formula (1+1). 2 marks for correct answer (1+1), e.g: • 2012 – $554/700 (1) \times 100 = 79\% (1)$ • 2013 – $562/830 (1) \times 100 = 68\% (1)$ NB if the answer given is 79% award 2 marks; if the answer given is 68% award 2 marks.	(4)

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 • Define subsidy. • Application – £554m – 79%/69% of income. Reasons for a subsidy include: • lowers production costs • increases supply (on diagram or written) • lowers price (on diagram or written) • increases quantity (on diagram or written) • more accessible to poor (low price) • higher meat and milk consumption • vital rural economy – protects incomes/jobs • maintains/guarantees food supply • trickle-down effect • food industry relies on them.	(9)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 Reasons against a subsidy include: • 79%/ 69% of income is subsidy • reliant on subsidy • X – inefficient • should be allowed to get out of market and make room for someone else • cost to government – opportunity cost.	(6)

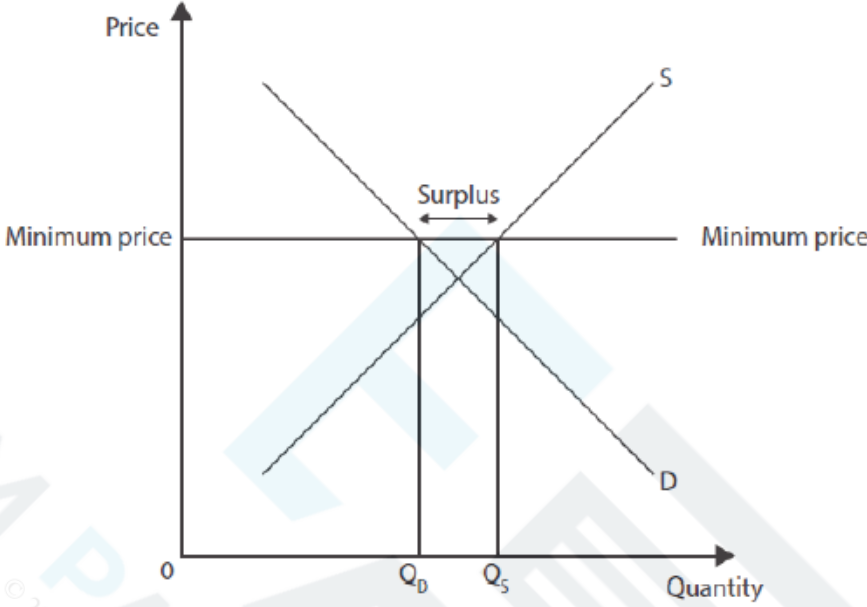
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Indicative content	Mark
(f)	Knowledge 4, Application 4, Analysis 6 - Define external costs. - Diagram accurately drawn – with MPC, MSB, MPB and MSC, welfare loss triangle.  - It takes 1,800 gallons of water to make a single pound of grain-fed beef. - Thirty per cent of the world's land is used to raise livestock. - Cows as major producers of methane gas. - These animals produce more climate-changing greenhouse gases than 22 million cars a year. - Manure, tons of fertilisers and toxic contaminants used in slaughterhouses.	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two-stage chain of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(f) continued	Evaluation 6 • Magnitude – depends on how much additional meat. • Measurement difficult – putting a value on external costs. • Increased demand – higher revenues. • Benefits of increased consumption – export income.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Question Number	Indicative content	Mark
(g)	Knowledge 4, Application 4, Analysis 6 - Diagram showing minimum price above market equilibrium.  - Define minimum price. - Reasons for minimum price – protect incomes of producers. - Raise price of beef/ cause excess supply. - Contraction of demand/ extension supply. - Government may have to buy excess supply - Reduce consumption of meat – reduce obesity. - Link to NHS.	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two-stage chains of reasoning only.
Level 3	7–10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	11–14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(g) continued	Evaluation 6 • Anti competition. • Magnitude of price changes – increase then may put businesses out of business – job losses. • Delays – time lag before it starts. • Encourage smuggling – if minimum price. • Imports. • Magnitude – depends on size of minimum/maximum price. • Elasticity of demand– if inelastic little change. • Time lag – people will not change behaviour immediately.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement.

Q9.

Question Number	Answer	Mark
	Knowledge 1, Application 2, Analysis 2 Knowledge/understanding 1 mark for e.g: - definition or formula or diagram of price elasticity of supply. Price elasticity of supply (PES) measures the responsiveness of quantity supplied to a change in price. PES can be calculated using the following formula: $\text{PES} = \frac{\text{percentage change in Quantity Supplied } (\% \Delta \text{ in } Q_s)}{\text{percentage change in Price } (\% \Delta \text{ in } P)}$ Application Linked application focussed on new housing stock (1+1) e.g. - Price elasticity of supply of 1 means unitary elasticity; $\text{PES} > 1$ means elastic; $\text{PES} < 1$ means inelastic (1) 1 mark for identifying Supply elasticities in the USA or Finland are at or above unity (1) and 1 mark for identifying a country with particularly low elasticities of supply e.g. the Netherlands or the UK (1). - Candidates who compare the elasticities of one country where supply is elastic and one country where supply is inelastic: e.g. USA +2.0 (1) and UK +0.4 (1). - Supply is particularly unresponsive to prices in the Netherlands, more than ten times less responsive than in the USA (1+1). - A numerical illustration of the effects of price changes in different countries (1+1)	
	Analysis Linked analysis (1+1) e.g. - Above unity PES implying that in response to a demand shock housing output will increase proportionally more than prices (1) and below unity where housing supply is relatively unresponsive (1). - Explaining how it is easier to build new houses in the US compared to the UK (1) where it is more difficult to build new houses e.g. due to planning restrictions in UK (1). - Allow other reasons/factors that affect PES e.g. land availability, technology or time (1+1).	(5)

Question Number	Indicative content	Mark
	Knowledge 2, Application 2, Analysis 2 • Help to Buy is a subsidy making home ownership more attractive • Cross (price) elasticity of demand (XED) measures the responsiveness of quantity demanded for one good to a change in the price of another good. • Identify home ownership and rented properties as substitutes • Likely positive cross elasticity of demand – Help to Buy scheme may result in decrease demand in rental sector • Deposit size 5% rather than 20% - demand increases • Demand for rental properties decrease – diagram showing shift inwards • Price and quantity of rental properties to fall • Consumer and Producer surplus in rented sector declines • Alternative views valid e.g. Help to Buy may increase house prices and therefore increase demand for rented accommodation from those not eligible for Help to Buy. Help to Buy shifted demand for private houses to the right (increase) and that, ceteris paribus, this would shift demand for rented accommodation to the left (reduction).	(6)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no links between causes and consequences
Level 2	3-4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relate them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5-6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
continued	Evaluation 4 • Scale – “modest” uptake of the Help to Buy scheme • Time-scheme due to end in 2017 • Scope – more likely to affect “lower end” of property market. • Closeness of substitutes – home ownership is purchasing an asset renting is a cost of living. • Scheme only affects first time buyers • PES – planning restrictions • Regional variations • Other factors more significant e.g. income • Candidates may refer to long-term investment in private rented sector as buy-to-let increasing demand for home ownership.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evaluative comments supported by chains of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Answer	Mark
	Knowledge 2, Application 2, Analysis 2 Knowledge 2 marks for e.g. - Define producer surplus e.g. a measure of producer welfare or the difference between what producers are willing and able to supply a good for and the price they actually receive - Diagram to show the original demand and supply curve and equilibrium - Identify the original producer surplus Analysis 2 marks for e.g. - Diagram illustrates increased producer surplus (1) resulting from demand shifting out (1) Application 2 marks for e.g. - Explain that there is an increase in prices (1) as demand for houses rises (1) - Explain that there is an extension in supply or a rise in the housing availability - House builders will make a larger surplus (profit) - Producer surplus is likely to be only for new housing. With the existing housing stock, any increase in house prices would not be referred to as increasing producer surplus as the seller still has to find a home to live in. - Effect of PED/PES on size of producer surplus	(6)
Question Number	Answer	Mark
	Knowledge 2, Application 2 1 mark for identifying the formula e.g. $(\text{new-original})/\text{original} \times 100$ (1 mark) 1 mark for locating the data 373 (Q1 2015) minus 300 (Q1 2009) 1 mark for performing calculation 1 mark for substituting figures correctly e.g. $(373 - 300)/300 \times 100 = 24\%$. Accept 23 – 25% NB If the answer given is 24% or 24 award 4 marks. Do not award inaccurate answers such as minus 24% as 4 marks but credit any accurate working.	(4)

Question Number	Indicative content		Mark
	Knowledge 3, Application 3, Analysis 3 - Prices perform a signalling function – they adjust to demonstrate where resources are required. - Prices act as an incentive function – for consumers and/or suppliers - Prices perform a rationing function. - House price index fell e.g. in 2007-09 from 360 to 300 (-17%). House price index increased e.g. 2009-2014 from 300 to 375 (+25%). - Use of selected PES in Figure 1 - Diagram to explain price mechanism e.g. removal of shortages and surpluses - Allow use of economic thinkers/theories e.g. Smith's invisible hand/addressing the economic problem, Hayek's communication network.		(9)
Level	Mark	Descriptor	
	0	A completely inaccurate response.	
Level 1	1-3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.	
Level 2	4-6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance	
Level 3	7-9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning	

Question Number	Indicative content	Mark
continued	Evaluation 6 • Signals ignore negative externalities of building on green belt or higher risk of flooding. • Incentives lack perfect information or exhibit asymmetric information • Housing is an asset – speculative demand will prevent shortages from being eradicated – a “runaway train” Extract A. • Housing price mechanism fails to meet demand for affordable housing. • Regional inconsistencies – supply shifting inwards in parts of country or demand not increasing • Impact of recession/housing recovery fragile - prices volatile or continuously over-priced. • Long run/ short run distinction. • Lack of land/planning restrictions means PES remains inelastic. • Allow other government interventions/failure.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	5-6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question.

Question Number	Indicative content	Mark
	Knowledge 4, Application 4, Analysis 6 • Effects include changing property prices, rental market, affordability, shifts in demand and supply, welfare changes, environmental impact, cost to firms, cost to government, or impact on other markets. • Methods of intervention – e.g. subsidies, taxation, provision of information, regulation, direct provision of housing, flood defences, Help to Buy and rent controls. One type of intervention is sufficient as long as considering range of micro-effects. • Purpose of Government intervention to correct market failure – housing shortages and affordability. Possibility of further market and government failure. • Use of Figure 1 to indicate UK's PES may increase. • Use of Extract B: to increase property supply "making new homes available more quickly". • Candidates may consider macro issues in a micro context if part of a chain of reasoning that identifies micro impact e.g. provides incentive to build more housing increases employment. NB KAA can be positive or negative effects.	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4-6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two-stage chain of reasoning only.
Level 3	7-10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance
Level 4	11-14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
continued	Evaluation 6 • Magnitude – extent of government intervention. • Measurement difficulty • Short run, long run issues. • Information gaps. • Unintended consequences - mortgage help may become ineffective in catching a 'runaway train' • Impact on different interest groups. • Distortion of price signals. • Opportunity cost to Government. • Environmental impact e.g. flooding NB EV can be positive or negative effects.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning
Level 3	5-6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement

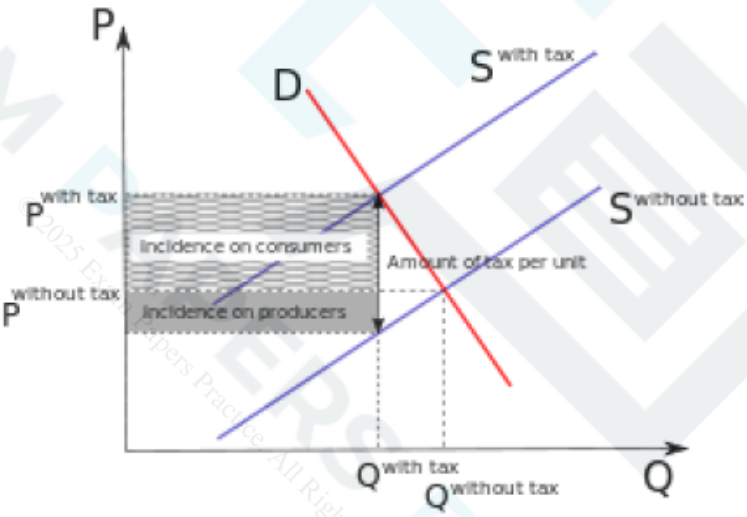
Question Number	Indicative content	Mark
	Knowledge 4, Application 4, Analysis 6 • Candidates may illustrate through use of appropriate diagram. • Public goods: non excludable and non rivalrous, e.g. flood defence schemes. • Failure of free market – free rider problem. • Externalities – positive or negative, e.g. housing may create these. • Information gaps: markets may have asymmetric information, e.g. insurance companies having incomplete information on risk of flooding. • Candidates may consider other types of government provision.	(14)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4-6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two-stage chain of reasoning only.
Level 3	7-10	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance
Level 4	11-14	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
continued	Evaluation 6 • Unintended consequences. • Magnitude • Government failure, e.g. flood defences. • Opportunity cost of government provision. • Ineffectiveness of government provision, e.g. climate change – better to abandon areas, • Information gaps. • Non-beneficiaries pay. • Anti-competitive.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning
Level 3	5-6	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation is balanced and considers the broad elements of the question, leading to a substantiated judgement

Q10.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2 Knowledge - Correct drawing of a demand and supply diagram annotated to show a parallel left (1) shift in the supply curve (1) Application - consumer tax incidence shown (1) - producer tax incidence shown (1)  NB: Award application mark for correctly identifying on diagram amount of tax per litre is set at 58 pence per litre. PED does not have to be price inelastic.	(4)

Question Number	Answer	Mark
(b)	Analysis 1 The only correct answer is A <i>B is not correct because both consumer surplus and producer surplus decrease</i> <i>C is not correct because producer surplus decreases</i> <i>D is not correct because consumer surplus decreases</i>	(1)

Q11.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2, Analysis 1 NB an answer which sees highly concentrated as having a large number of firms will not earn knowledge or analysis marks Knowledge and analysis: 3 marks for e.g. • Identification of high concentration - high market share of some firms (1) few large firms dominate (1) uncompetitive (1) • Identification of market structure - there is a <i>legal</i> monopoly (Walkers) OR oligopoly OR duopoly (1) • Firms can set prices (1) control output (1) / monopoly power (1) • Interdependent (1) • High barriers to entry/exit (1) low contestability (1) Application: 2 marks for data references e.g. • Walkers have over 50% of market share (1) • Use of concentration ratio e.g. 2 firm CR 78.0% (2), 3 firm CR 82.2% (2) • Extract A refers to ban making market more competitive (1) implying prices are currently uncompetitive/high (1) • Extract A – high levels of advertising (1) • common features: they tend to be dominated by a small number of firms that sell multiple brands and that heavily advertise their products (1) NB Concentration ratio calculation can be awarded as analysis or application Award a maximum of 1 application mark if there is no reference to Figure 1 or Extract A	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge/ implicit understanding: (1+1) marks for reason e.g. • Educational/skills/wage differences • Changing patterns of trade/globalisation • Government policy/austerity/universal credit • Income generated from inheritance • Payments generated from debt • Race, gender, age, geographical issues • Changes in direct tax • Stage of the economic cycle/Financial Crisis 2008 • House prices (must relate to income not wealth) • Weakened power of trade unions • Level of unemployment/underemployment • Irrational behaviour/information gaps/myopia Application (1+1 or 2): • Any valid application to causing inequality in the UK or other developed country, e.g. unaffordable house prices in the UK mean the younger generation has little income after rent (1); cut in the top rate of income tax, rise in NICs, change in tax thresholds; university fees at £9250 a year is widening the income gap if fewer low income households go to university (1); pensions rising below average incomes (1); educational differences such as an economics degree has a higher impact on future earnings than other degrees (1)	
	Analysis (1+1): • Linked development of the two points made Evaluation (1+1 or 2). Points might include discussion of: • The role of transfer payments/minimum wages in narrowing/failing to narrow inequality • Changes rather than levels of income inequality, absolute vs relative issues • Relationship with wealth e.g. as a more significant inequality issue • Relative significance of different factors e.g. direct taxes more proportional than indirect taxes (regressive) • Time lags e.g. education has effects as yet unfolded, government policy might change, Brexit has not yet happened. NB Do not allow indirect taxes or subsidies as KAA NB Use of Gini coefficient or Lorenz curves could be used as application or analysis	(8)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 - Firms are interdependent so they will respond in the face of what other firms might do - Firms may not reach the same outcome as if they operated alone - Firms are cutting prices because they want to expand market share/steal other firms' markets - All firms responding in this way means that firms make less profit overall - Effects of advertising ban may make market more contestable, with possible effects on entrant - Price cuts could be a form of limit pricing, i.e. reducing contestability (effects on possible entrants may be shown) Game theory e.g. a payoff matrix or kinked demand curve to illustrate behaviour, for example cartel-like or collusive behaviour, price leadership NB to access Level 3 there must be correct use of game theory and application to the data.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 - Firms such as Walkers have significant market share and brand loyalty, even if competition lower price they may not switch - Use of context to show that firms are cutting prices because of a regulation and not because of the actions of other firms. No sign of interdependence. - Size of market shrinking overall so behaviour more aggressive - Firms setting low prices in a long run equilibrium might not be the best option for the firms or other stakeholders. - Advertising ban affects children mainly - so price competition less needed for older target audiences that can still be targeted. - Reference to only 10% loss in sales compared to 15% - still a large impact on firms (Extract A).	(4)
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 16 marks for KAA, for effects of increased spending on education to promote healthy eating in the UK. Microeconomic effects may include: • Effects of increased demand for healthy foods/consumer surplus/increased prices (may include supply and demand diagram) or fall in price of HFSS foods • Positive externalities in consumption are internalised (may include divergent MSB and MPB diagram and corresponding movement toward social optimum at $MSB=MSC$) • Impact on firms which make unhealthy products – increased costs if they diversify/reduced demand e.g. use of healthier ingredients with higher costs. This might be shown on a cost/revenue diagram. • Increased profits for healthy food producers e.g. shown on a costs/revenue diagram • labour market issues, e.g. workforce becomes healthier, and reduced obesity might make workers more productive (increase marginal product) • Opportunity cost of spending on education Macroeconomic effects may include: • Effects on price level and output as G increases, with multiplier effects. May be shown by shift out in AD with effects on price level and real output. • Healthier workforce shifts out the LRAS, with impact of lower prices and higher output • Fiscal implications – crowding out, effect on balancing budget	
	NB for a Level 4 response there must be micro and macro effect (s). Opportunity cost/shift in PPF/labour markets/competitiveness etc. can be used as micro or macro. 9 marks for evaluation – questioning might include: • Expansionary fiscal policies not necessarily happening. It might mean expenditure elsewhere is cut (opportunity cost arguments) • Trade-off between objectives e.g. increased spending might cause inflation, depends on the state of the fiscal balance, multiplier etc. • Questioning whether education is effective, or whether other policies are needed, e.g. tax, banning advertisements • Education might be counter-productive/role of state might be questioned e.g. government failure, crowding out effect	(25)

Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5–8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9–12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13–16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic factors. The answer demonstrates logical and coherent chains of reasoning.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8 Evaluation 9 16 marks for KAA, for effects of increase in tax on HFSS foods. Microeconomic effects may include: • Rational economics implies people will change consumption patterns. Consumer preferences will change away from HFSS foods e.g. as consumers switched to cheaper brands and crossed the border to Sweden and Germany to do their shopping (Extract C). • An ad valorem or specific tax diagram might be used to illustrate the effects on prices/quantity in each market, and the incidence of tax • Cross elasticity of demand – evidence (e.g. Extract C) that a rise in price may cause a rise in demand for healthy food. • Welfare changes, e.g. loss of consumer, producer surplus/higher prices and deadweight welfare loss. This might be shown with an externalities diagram, e.g. positive in consumption of healthier foods. • Impact on firms, e.g. loss in profits on HFSS as costs rise/demand falls. This might be illustrated using a costs/revenue diagram; increased profits for healthier options. • Impact on labour market. Healthier workers, or effects of cut in real wages as costs of living increase. Changing employment opportunities. e.g. reduced employment for HFSS manufacturers and increased employment for healthy foods firms.	
	Macroeconomic effects may include: • Redistribution (micro or macro issue): indirect taxes of this sort are regressive, disproportionately affecting the elderly and the poor. Taxes on HFSS foods moderate consumers, as well as the obese overeaters they are intended to target. • Cost push inflation – a one-off rise in prices. This might be shown by an upwards/leftwards movement of AS • Effects on incomes/net exports – may be linked to a fall in AD, e.g. in Denmark the policy helped push up food prices in a year in which real wages fell by 0.8%. • Governments might gain revenue which can be used elsewhere in the economy / lose revenue – Laffer curve (Extract C) NB for a Level 4 response there must be micro and macro effect (s). Opportunity cost/shift in PPF/labour markets/productivity/competitiveness etc. can be	

	used as micro or macro. Up to 9 marks for evaluation. Points might include: • Depends if firms respond by cutting prices (Extract B) • The tax makes people buy other unhealthier food (in Extract C) e.g. increases price and people may substitute to cheaper even unhealthier or low-quality alternatives that are more affordable • Tax is not an efficient deterrent to obesity. The policy had a very limited impact on the consumption of HFSS in the data foods: 80% of Danes did not change their shopping habits at all (Ext C). Not only is demand for food relatively inelastic, prices often fluctuate, and consumers become immune to changes • Depends on magnitude- the proportion of income may be low • The price elasticity of demand will determine the incidence of the tax - inelastic larger consumer incidence and elastic larger producer incidence. • Depends on how funds are used - e.g. to promote healthy eating through subsidies and education • The intervention may mean free markets work less well – theory of unintended consequences • Other unintended consequences e.g. counterfeit, smuggling	(25)
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Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5–8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9–12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13–16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic factors. The answer demonstrates logical and coherent chains of reasoning.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Q12.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 1, Analysis 3 Knowledge/Understanding: (up to 1 mark) • Real price of gas and electricity refers to their price adjusted to remove the effects of inflation (1) • Increase in demand (1) • Decrease in supply. (1) • The lack of competition (1) Application: (up to 1 mark) e.g. • Figure 1 indicates the real price of gas and electricity has increased since 2004: electricity from 70 to 119 or rises 70% (accept approximations) (1) • Gas from 58 to 121 or rises by 108% (accept approximations) (1) Analysis: up to 3 marks for linked development of one reason e.g. • Rise in price (1) due to non price determinant of demand e.g. income, economic growth, population growth (1) explained using contraction in demand or extension in supply or other extended analysis of market forces (1) • Rise in price (1) due to non price determinant of supply e.g. increase in cost of production, finite resources (1) explained using contraction in demand or extension in supply or other extended analysis of market forces (1) • Energy firms have sought to increase profits (1) tacit collusion (1) provide more dividends and higher share prices for shareholders (1) directors salaries linked to profits (1) • Energy firms able to exploit consumers not behaving rationally due to: influence of other consumers (1) habitual behaviour (1) computation difficulties (1)	(5)

Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 4 • Measures referred to will increase competitiveness by removing imperfect information, addressing asymmetric information, reducing barriers to entry and exit. • Customer database will help producers to directly contact consumers to offer better deals. <i>It may encourage firms to enter the market – the extract refers to recent new entrants having a combined market share of 12% for gas and electricity. Firms may enter as they have more market knowledge of customers for each energy supplier and are permitted to contact them directly offering their service.</i> • Smart meters will increase market information for consumers on their electricity consumption so may seek better deals. <i>Scope exists to enter as 70% of domestic customers of the six largest energy firms on an expensive standard tariff. Smart meters provide more information to customers on energy usage and price / reduce issue of asymmetric information / forcing energy suppliers to become more price competitive.</i> • Price cap for vulnerable customers on pre-paid meters will help limit increases in energy bills and so offer some protection as a surrogate for competition. <i>Price cap may offer some degree of certainty over the price firms could charge / encouraging market entry.</i> • Other evidence from Extract A demonstrates that measures have been successful in the past e.g. new entrants in the market NB: Answers must relate to effectiveness of measures NB: Arguments that the measures are successful may be seen as KAA and unsuccessful as evaluation or vice versa	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(b) continued	Evaluation 4 - Some consumers may still not switch to cheaper energy tariffs / due to problem of computation, inertia or habit. - Customer database may lead to unintended consequences e.g. customer annoyance at telephone calls so they ignore calls, information overload resulting in making it harder to compute to make a decision. - Significant barriers to entry exist in the domestic energy market: large economies of scale/established firms have some customer loyalty / advertising costs / licences required to supply energy. - Price cap is only temporary and covers just 4 million customers / energy prices through pre-paid meters already exceed standard bills. - Price cap may discourage firms from entering the market / as less profit incentive. - Price cap may automatically become the price each company sets and so reduces competition and may increase prices. - CMA found no evidence of anti-competitive practice.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 2 - Understanding of profit cap (a limit on profits as a percentage of total revenue). Linked development: profit cap is usually introduced in industries where supernormal profits are regarded as excessive / profit cap may be necessary where little competition exists or where collusion is possible. - Chair of CMA recommended 1.25% profit cap against industry return of 7%. Linked development: profit cap is proposed in the retail sector of the energy industry which involves just marketing, billing and metering energy. - Impact on consumers: lower energy bills and so households may be better off in real terms / may increase energy consumption / domestic firms that consume energy may become more competitive due to lower costs of production / rising consumer surplus. - Impact on energy producers: lower revenue and so lower profits / less funds for investment and so risk of energy supplies in future / falling share prices and dividends / change in objective e.g. from profit maximisation to sales maximisation. NB: For Level 3 response candidates must refer to consumers and supplies NB: Negative effects on consumers and producers may be seen as KAA and positive effects as evaluation or vice versa	(6)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 • There may be upward effects on price for consumers as firms become more x-inefficient or reduce total investment. • Depends on the level of profit cap on energy firms / how long it may be applied: if it is 1.25% it may be too harsh compared to other industries such as supermarkets. • It is a vertically integrated industry – so energy companies can make profits in the generation of electricity and gas. • Energy firms appear to have made huge profits over previous years and so may be able to cope with lower profits. • Energy firms may increase sales revenue and so profit values may still be high. • Energy suppliers experience low levels of risk in their retail operations so necessary rewards are lower. • Unintended consequences such as a reduction in new entrants to the industry and less competition, risk of power shortages due to under investment or excess demand. • Multinational owners may use transfer pricing to avoid profit cap.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Answer	Mark
(d)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Accept answers based on switching between electricity suppliers as well as switching from electricity to other energy sources Knowledge/Understanding: 2 marks for identification of two reasons (1+1) e.g. • It takes time for consumers to switch demand following price changes • Existing contracts make it difficult to switch in the short term • More substitutes available in the longer term Analysis: 1 mark for linked explanation of each reason (1+1) e.g. • Demand more price inelastic in short run as: takes time for consumers to adjust their consumption habits (habitual behaviour) / imperfect market information / no close substitutes. • Demand less price inelastic in long run as: some consumers are able to change their habits improved market information / substitutes may emerge such as greater use of gas appliances or installation of solar power panels.	

	Application: 2 marks for reference to data (1+1), e.g. • Idea that demand is price inelastic (1) becoming less inelastic / more elastic in the long run (1) • Use of Extract A e.g. 70% of domestic consumers do not switch (1) • Development of new technology that reduces electricity consumption / electricity saving lightbulbs or energy efficient kettles (1) • Increased awareness of measures to become more price sensitive (1) to become more energy efficient (1) • Costs and time involved in switching to energy-saving appliances (1)	
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	• Improved consumer knowledge of consumption through smart meters (1) • New firms might enter the market using gas / biofuel / solar / wind / households producing their own rather than buying electricity (1) Evaluation: 2 marks for two evaluative comments, OR 2 marks for identification and linked development of one comment • The figures are averages for all households – but great variation exists between households; for example, there may be no effect on households who have little concern for energy saving measures. • High income households may not be affected by rising electricity prices as it comprises such a small part of total income - so demand remains very price inelastic. • Some households may suffer from imperfect market knowledge in long run or cannot change their habits – so demand remains very price inelastic. • Consideration of the magnitude of the change in elasticity (from -0.35 to -0.85); quite substantial in terms of doubling elasticity / or electricity demand still remains price inelastic in the long run as no suitable alternatives.	(8)
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Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 Policies may include: • Increase investment in training programmes to raise quantity and quality of human capital and productivity / increase provision of modern apprenticeship schemes / it may involve government subsidies or tax breaks to firms. • Recruit skilled workers from overseas via careers fairs and trade journals / provides a ready supply of labour. Accept policies referring to geographical mobility of labour if linked to the energy sector. • Change industry image so it appeals to more female school leavers and graduates – this may involve restructuring of careers advice / marketing programmes and information on career paths and opportunities. • Retain ageing workforce by improving working conditions and rates of pay / offer flexible working hours and part-time work / raise state retirement age; older workers have higher levels of human capital and so productivity could be increased. NB: For Level 3 response candidates must refer to at least one business policy and one government policy. NB: For Level 3 response candidates must refer to the energy sector.	(9)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • Training programmes involve (opportunity) costs to businesses and government / time period to train labour, especially in higher order technical skills / quality of training programmes. • Recruitment of skilled workers from overseas may create problems such as access to UK jobs following Brexit decision / language barriers / differences in qualifications and skills to those required / possibility of higher staff turnover if overseas workers intend to return home / family ties and high house prices may create geographical immobility of labour. • Changing energy sector image may require high levels of marketing expenditure / some jobs may not suit parents with family commitments. • Retain older workers could lead to problems of poor health and low productivity and ability to learn new tasks. Also the underlying skilled labour shortage remains – eventually older workers will retire. • Other policies to overcome labour shortage for example, new technology to substitute labour with capital, means that reducing labour immobility is not a priority.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Q13.

Question Number		Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge/understanding and analysis: Oligopoly (1) OR Legal monopoly (1) OR Monopoly only if monopoly POWER is implied e.g. using CR calculation (1) OR Duopoly only if duopoly POWER is implied e.g. using CR calculation (1) AND Understanding of market structure identified, e.g. a market <i>dominated by a few firms</i>, or <i>interdependent</i> (1) Explanation of the process linking to the market structure to the action of firms e.g. high concentration means that firms may exhibit collusion, price rigidity, use of non-price competition, profits are not competed away (1)	
	Application (1+1): 2 marks for data references from Figure 1 and other sources (at least 1 mark must be related to Figure 1) e.g. Others 29% - Concentration ratio calculation e.g. 2 firm CR is 44% 3 firm CR is 53% 4 firm CR is 60% 5 firm CR is 66% 6 firm CR is 71% (2) - Costa (31%) is >25% market share (2) - Strong brand names, brand logos e.g. Caffè Nero and Costa (1) have been dominating the UK market Fig. 1/Extract A (1) - Market power is likely to be high (1) with reason e.g. consumer loyalty (1) shown in Figure 3 as price differences tolerates/illustrate market power (1) coffee has price inelastic demand (1) so people are prepared to pay more (1)	(5)

Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 4 Pricing strategies - must be linked to <i>profit</i>, and may include: • Price leadership/price matching – some brands seem very similar in pricing e.g. use kinked demand theory • Predatory pricing • Limit pricing (with view to higher future profits) • Price wars (with view to higher future profits) • Collusion/collaboration - reward the use of game theory/pay off matrix to support this • Undercutting close rival e.g. Costa (with view to higher profits) • Also award, if linked to profit: sales max, exploiting economies of scale, cost cutting, or competitive pricing linked to elastic demand, price rises and/or price discrimination	
	Non-pricing strategies – linked to <i>profit</i>, might include: • Loyalty schemes, e.g. Starbucks only allow its own cup for reuse in store, • Paying for central and expensive locations for stores • Opening up new stores near to competitors to undermine the competition, e.g. at a London location there are many competitors • Closing down non-profitable stores as rationalisation policy • Strengthening the brand by advertising • improving reputation for recycling washable cups/fair trade coffee, sharing with other brands not just Starbucks coffee cups which can be washed in store and re-used • Extension of product range • Focus on quality NB Award collusion and BOGOF-style approaches as either pricing or non-pricing strategies. NB For a Level 3 answer, there must be both price and non-price strategies, and reference to the context of <i>profitability of Starbucks</i>.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(b) continued	Evaluation 4 • Weighing up of reasons for using price or non-price competition, e.g. • Not likely to use price competition - e.g. better to collude and keep higher joint profits. • Non-price competition can have very uncertain results/be expensive • Effects in the short run and long run might vary e.g. rising costs lead to falling profits, but long run demand rises leading to better results • Relative effectiveness of measures e.g. limit pricing may be static in long run and predatory pricing changes when other firms leave • Some of these strategies are illegal and could lead to fines, which might reduce overall profitability • Unstable nature of game theory equilibria • The effectiveness of the competition authorities might be considered e.g. regulatory capture • Depends on the strength of brand loyalty • Limitations of loyalty schemes, e.g. cost, consumer behaviour.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number		Mark
(c)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge (1+1) e.g. • Fall in consumption of non-renewable products (1) • Rise in tax revenues (1) • Reduction in external costs/greater environmental awareness (1) • Soft touch regulation/allows consumer choice (1) Application (2 or 1+1) e.g. • Use of reusable cups as a means of reducing demand for disposable cups (i.e. a substitute, Extract C) (1) • Extract B – 5p charge is a small proportion of £2.60 (1) • 85% reduction in plastic bags (1) • This might be illustrated through the use of a correct supply curve shift, e.g. a fall in consumption or production of cups on x axis (1) Analysis (1+1) e.g. • Indirect taxes may raise prices (1) • Increases in costs of production (1) • Link between unit tax, tax revenue and reduction in demand for cups (1) • A supply and demand diagram showing an indirect tax (supply shift upwards/leftwards) or an externalities diagram might be used (1) • tax revenue could be used to subsidise other products, e.g. recyclable cups (1)	

	Evaluation (2 marks for any relevant point or 1+1). Points might include: • It depends if the tax is passed on to the consumer • Other policies may be considered e.g. new recycling specialist facilities, regulation • PED might be used, e.g. a 5p tax on the price of a £2.60 coffee might have very little effect on demand, or, a cup of coffee may be a small percentage of consumer income, depends on the availability of substitutes • Regressive nature of the tax – has a greater effect on lower incomes • Other approaches considered, e.g. subsidising reusable cups as in Ext B changing the technology in the recycling NB award two advantages. A diagram can be credited as analysis or application (1+1) up to 2 marks in total.	(8)
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Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic influences may include: • Level of demand or expected demand, prospect for profit • Cost of labour, minimum wage, spare capacity in the labour market • Rent/land prices • Contestability of market/barriers to entry e.g. how long and it takes and ease of setting up business • Legal system and other legislative issues • Property rights • Extent of regulation • Existing PED • Brand loyalty • Sunk costs Macroeconomic influences may include: • Levels of competitiveness e.g. price and non-price, which might relate to competitiveness ranking, e.g. infrastructure, human capital, institutions • Macroeconomic environment, e.g. inflation • Membership of EU or other trade bloc issues • Value of currency • Level of corporation tax. • Transfer pricing possibility, as outlined in Extract A, as impact on profitability (not costs) • Level/rate of national income/economic growth and spare capacity/unemployment • Growth of household incomes in different countries, e.g. Chinese coffee shops are the most profitable, growing quickly in number • Strength and stability of local economy • Political and social stability	

	NB competitiveness, corporation tax, incomes, unemployment and contestability can be seen as a micro or macro approach. - Possible evaluation points include: - Factors might balance each other out - Changes in tax rates, e.g. government cutting corporation tax significantly - Starbucks is starting to report UK profits – perhaps it has had too much negative publicity from the low profit tax paid, Extract A reduction by a third in sales at Starbucks. Also it is moving headquarters from Netherlands to UK - Different influences affect different firms	(25)
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Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5–8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9–12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13–16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic factors. The answer demonstrates logical and coherent chains of reasoning.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evidence of evaluation of alternative approaches which is unbalanced, leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.



Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic effects may include: • Effects on consumers – e.g. higher prices (or lower prices if increased competition), more consumer spending on high street. Might include PED, YED (this might be developed as evaluation) • Effect on workers – higher incomes, more jobs • Effect on firms, e.g. increase in profits, revenue/decrease for independent non-branded coffee shops; cost of employing specific skilled workers e.g. baristas or other factors of production • Increase negative externalities e.g. more coffee cups • External economies of scale e.g. ancillary services such as cup manufacturers Macroeconomic effects may include: • Increased employment in service industry • Balance of payments e.g. outflow of profits to Starbucks • Effects on primary product producers • AD increase, negative/positive multiplier effects (local businesses might lose out to branded supermarket coffee outlets), inflation/growth • Tax implications e.g. transfer pricing • Income inequality • Environmental impacts	
	NB environmental impacts, labour markets, taxation or incomes can be seen as a micro or macro approach. Evaluation points may include: • Extent of negative externalities depends on whether new processing plants for cups are set up/how alternatives are produced. • Depends on the extent of the increase in demand, PED, YED, XED – and if other areas are seeing declines. • Depends on the size of the multiplier • Depends on the level of spare capacity (LRAS for example) • Depends on growth levels, point at which AD and AS meet (if Keynesian approach) • Depends on the performance of other firms • Depends on the extent to which local shops are displaced • Depends on the extent to which substitution can occur – e.g. people drink coffee instead of going to pubs • Depends on degree of profit repatriation, remittances, transfer pricing • Depends on whether coffee shops extends consumption spending or replaces demand for something else which might have different benefits/costs	(25)

Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5–8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9–12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13–16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic factors. The answer demonstrates logical and coherent chains of reasoning.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evidence of evaluation of alternative approaches which is unbalanced, leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Q14.

Question Number	Answer	Mark
(a)	The only correct answer is C <i>A is not correct because OP_1HQ_1 is the total cost of the product plus tax revenue producers' share of the tax</i> <i>B is not correct because P_2P_1HJ is the consumers' share of the tax</i> <i>D is not correct because P_3P_2JK is the producers' share of the tax</i>	(1)

Question Number	Answer	Mark
(b)	Knowledge 1, Application 1, Analysis 1 Knowledge/understanding 1 mark for correct definition of specific tax e.g. specific tax is a set amount per unit of the product sold Application 1 mark for e.g. of specific tax such as: • fuel duty • alcohol and tobacco duty • gambling duties • air passenger duty • sugar tax Analysis 1 mark for linked development e.g. to internalise the externalities accruing to the consumption of products causing external costs (1). Reference to 'sin taxes' (1). Raising revenue (1).	(3)

Q15.

Question Number	Answer	Mark
(a)	Application 2 Application 2: 1 mark for appropriate calculation /formula e.g. $\frac{1.3 \times \pounds 489\,000}{100} \quad (1)$ OR $489\,000 \times 0.013 \quad (1)$ Answer = £6 357 Award 2 marks for correct answer (6 360)	(2)

Question Number	Answer	Mark
(b)	The only correct answer is C <i>A is not correct because the candidate has mistakenly done the average house price in North East of England to the average house price in London</i> <i>B is not correct because the candidate has made a computational error</i> <i>D is not correct because the candidate has made a computational error</i>	(1)

Question Number	Answer	Mark
(c)	Knowledge 1, Analysis 1 Knowledge / understanding: 1 mark for one reason e.g. • differences in average income or wealth (1) • employment or unemployment rates (1) • migration or population change (1) • foreign buyers (1) • availability of land to build on (1) Analysis: 1 mark for a linked development e.g. • Higher average income in London mean people are able to obtain larger mortgages and so pay higher prices (1) • Higher employment rate or lower unemployment rate in London mean greater demand for housing from its population (1) • Foreign buyers attracted to invest in London due to high rental returns or expectations of increase in value of property or better market knowledge (1) • Shortage of land to build on in London, especially with green belt restrictions (1)	(2)